UNIVERSITY OF WESTERN ATTICA

DEPARTMENT OF INFORMATION AND COMPUTER ENGINEERING

DIGITAL DESIGN

WORKSHOP 3
FLIP-FLOPS

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CONTENTS

CHAPTER 1

A FEW WORDS ABOUT THE WORK

The work is based on the " Flip-Flop " circuits and their operation. In more detail, it is based on the sequential circuits, the " NAND ", " NOR " gate latches, the " RS Flip-Flop ", the " D Flip-Flop ", the " JK Flip-Flop ", the " T Flip-Flop ", the " Master-slave Flip-Flop ", the excitation of the " Flip-Flop " and the asynchronous inputs.

CHAPTER 2

BIBLIOGRAPHY

- Design and implementation of logic circuits P
. Giannakopoulos L. Aslanoglou

CHAPTER 3

DESCRIPTION OF THE IMPLEMENTATION OF THE WORK

For the implementation of the project the simulator " multisim " was used for the design of the circuits. In more detail, cables, grounds, VCC sources, lamps, logic gates (AND , OR , NOR , NAND , XOR , XNOR , NOT), oscilloscope were used.

CHAPTER 4

EXERCISES

EXERCISE 5.2.1 NAND GATE PEGGING EXERCISE 5.2.1

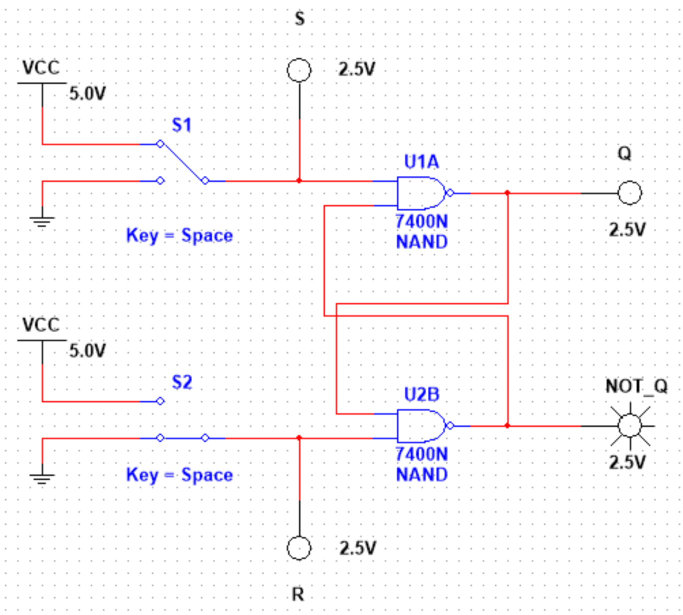
THEORETICAL RESULTS

The truth table of the pegged " NAND gate" pegger is as follows:

S	R	Q	NOT_Q
0	0	0	1
1	0	1	1
0	1	0	1
1	1	1	0

ADVANCE NOTIFICATION

For S = 0 and R = 0



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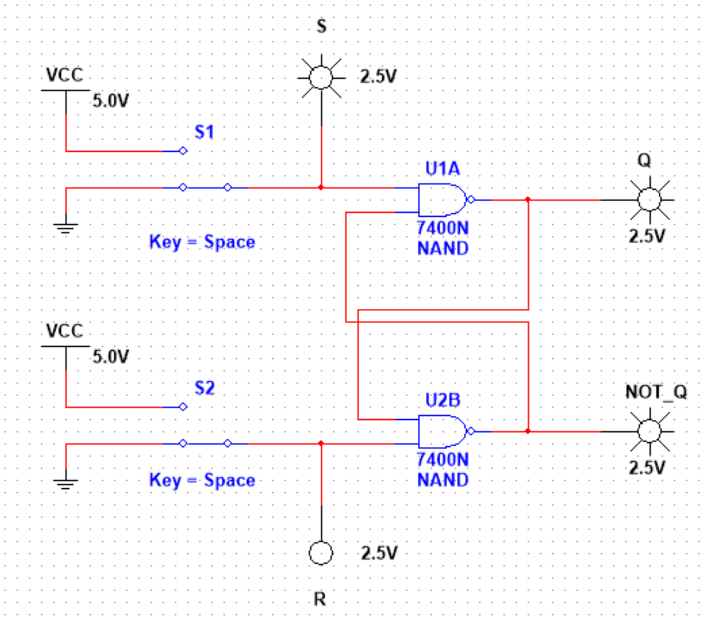

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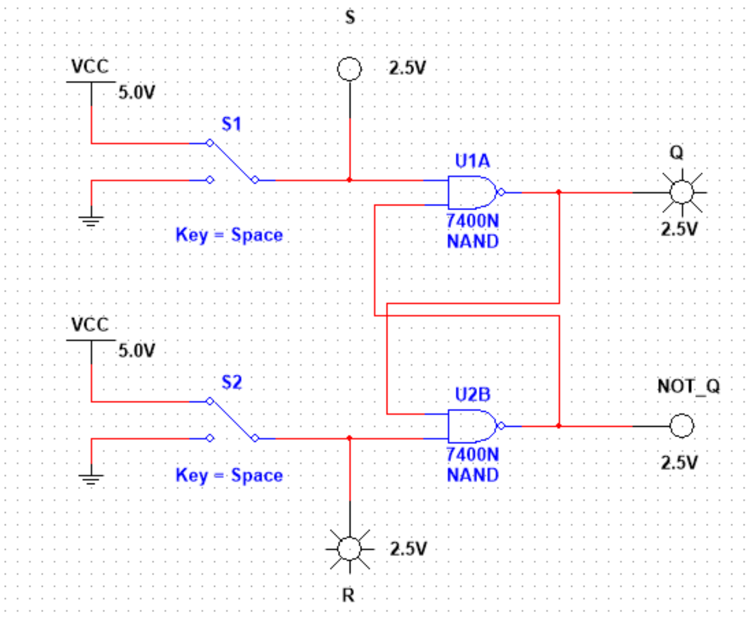

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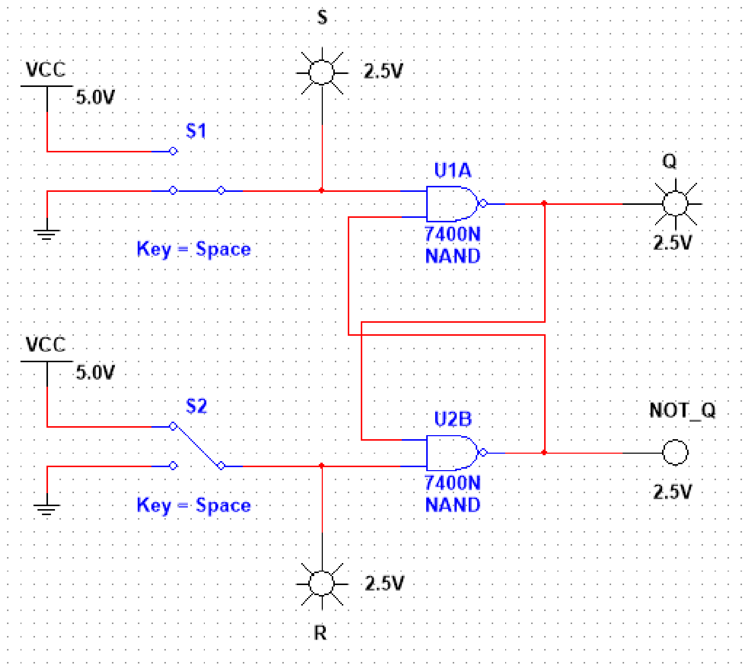


EXERCISE 5.2.2.2 RS FLIP-FLOP

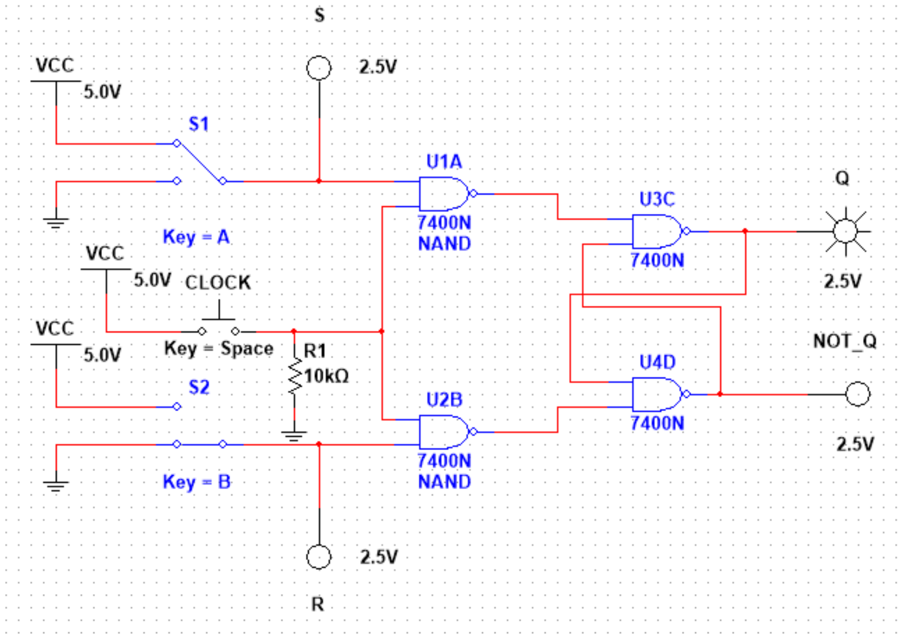
THEORETICAL RESULTS

The truth table of the RS Flip-Flop is as follows:

S	R	Q	NOT_Q
0	0	1	0
1	0	?	?
0	1	1	1
1	1	0	1

ADVERTISEMENT

For S = 0 and R = 0



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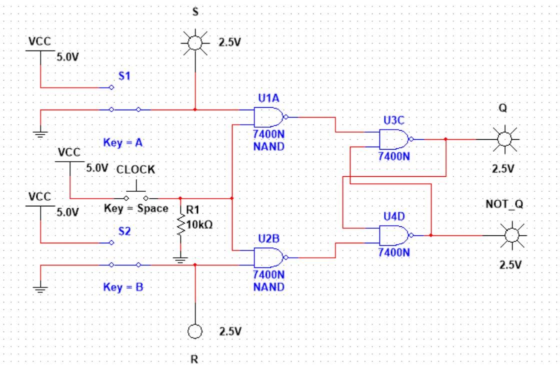

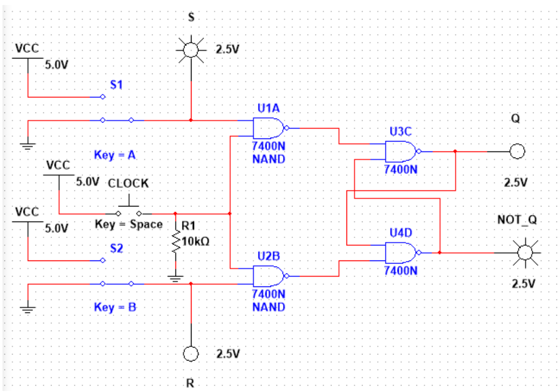

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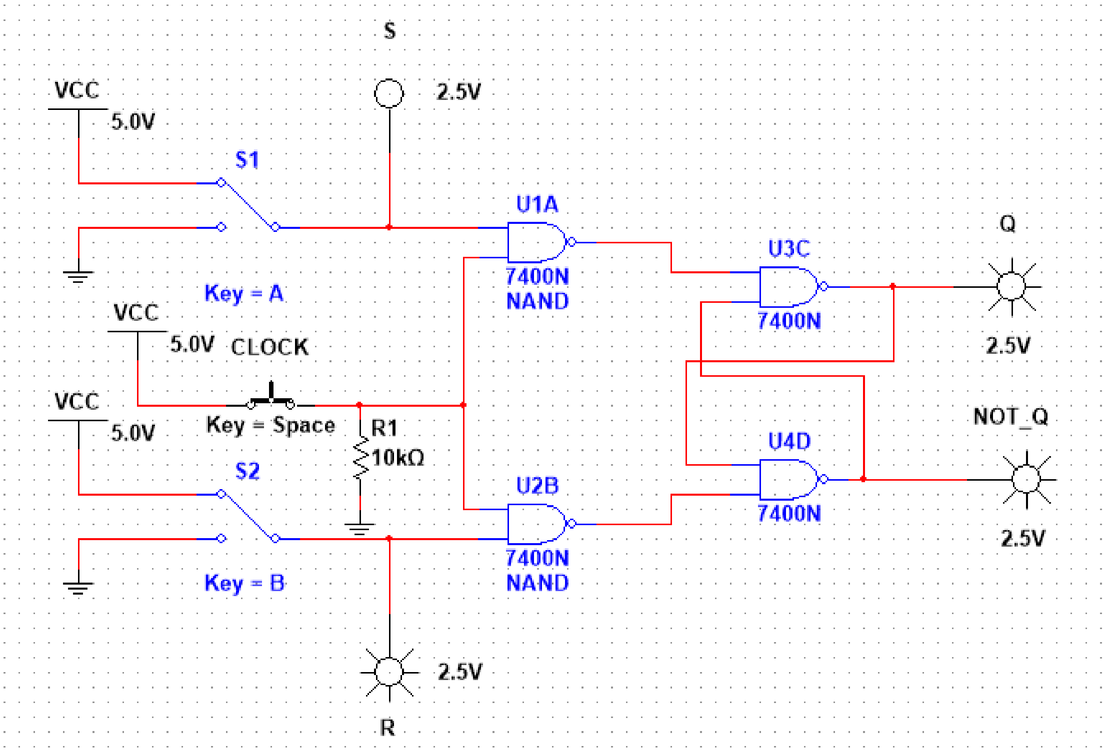

For S = 1 and R = 1